

$$\begin{array}{r}
 12345678 \mid 16 \\
 -12345664 \mid 771604 \mid 16 \\
 \hline
 121 \quad 771600 \mid 48225 \mid 16 \\
 \hline
 4 \quad 48224 \mid 3014 \mid 16 \\
 \hline
 1 \quad 3008 \mid 188 \mid 16 \\
 \hline
 6 \quad 176 \mid 11 \mid 16 \\
 \hline
 12 \quad 0 \mid 0 \\
 \hline
 11
 \end{array}$$

Ответ: BC614E

$$\begin{array}{r} 1000000 \mid 16 \\ -1000000 \mid 62500 \mid 16 \\ \hline 0 \mid 62496 \mid 3906 \mid 16 \\ \hline 4 \mid 3904 \mid 244 \mid 16 \\ \hline 2 \mid 240 \mid 15 \mid 16 \\ \hline 4 \mid 0 \mid 0 \\ \hline 15 \end{array}$$

Ответ: F24240

Задание 2.1 Перевести из 16 в 10 систему 12345678

$$\begin{aligned}12345678 &= 1 \cdot 16^7 + 2 \cdot 16^6 + 3 \cdot 16^5 + 4 \cdot 16^4 + 5 \cdot 16^3 + 6 \cdot 16^2 + 7 \cdot 16^1 + 8 \cdot 16^0 = \\&= 268435456 + 33554432 + 3145728 + 262144 + 20480 + 1536 + 112 + 8 = \\&= 305419869\end{aligned}$$

Ответ: 305419869

Задание 2.2 Перевести из 16 в 10 систему 1000000

$$\begin{aligned}1000000 &= 1 \cdot 16^6 + 0 \cdot 16^5 + 0 \cdot 16^4 + 0 \cdot 16^3 + 0 \cdot 16^2 + 0 \cdot 16^1 + 0 \cdot 16^0 = \\&= 16777216\end{aligned}$$

Ответ: 16777216

Задание 3

Записать в виде логического выражения ответ Винни Пуха: «Сгущённого молока и мёда и можно без хлеба»

Дать сгущенного молока – S;

Дать меда – M;

Дать хлеба – X;

Соответственно результат $R = S \ \&\& \ M \ \&\&!X$;

Ответ: $S \ \&\& \ M \ \&\&!X$;

Задание 4 Доказать тождества $A \rightarrow B = \neg A \vee B$ и $A \leftrightarrow B = (A \wedge B) \vee (\neg A \wedge \neg B)$ (таблицы истинности на Си можно распечатать)

```
#include <stdio.h>

int main(void) {
    int not_a = 0, not_b = 0;
    printf("A -> B = !A || B\n\r");
    printf("\n");
    printf("A | B | A -> B\n\r");
    printf("0 | 0 | 1\n\r");
    printf("0 | 1 | 1\n\r");
    printf("1 | 0 | 0\n\r");
    printf("1 | 1 | 1\n\r");
    printf("\n");
    printf("A | B | !A | !A || B\n\r");
    for(int a = 0; a <= 1; a++)
        for(int b = 0; b <= 1; b++)
        {
            not_a = !a;
            printf("%d | %d | %d | %d\n\r", a, b, not_a, not_a || b);
        }
    printf("\n");
    printf("A | B | !A || B | A -> B\n\r");
    printf("0 | 0 | 1 | 1 | 1\n\r");
    printf("0 | 1 | 1 | 1 | 1\n\r");
    printf("1 | 0 | 0 | 0 | 0\n\r");
    printf("1 | 1 | 1 | 1 | 1\n\r");
    printf("\n");
    printf("A <-> B = (A && B) || (!A && !B)\n\r");
    printf("\n");
    printf("A | B | A <-> B\n\r");
    printf("0 | 0 | 1\n\r");
    printf("0 | 1 | 0\n\r");
    printf("1 | 0 | 0\n\r");
    printf("1 | 1 | 1\n\r");
    printf("\n");
    printf("A | B | !A | !B | A&&B | !A&&!B | (A && B) || (!A && !B)\n\r");
    for(int a = 0; a <= 1; a++)
        for(int b = 0; b <= 1; b++)
        {
            not_a = !a;
            not_b = !b;

            printf("%d | %d | %d | %d | %d | %d | %d\n\r", a, b, not_a, not_b, a&&b, not_a&&not_b, (a&&b) || (not_a&&not_b));
        }
    printf("\n");
    printf("A | B | A <-> B | (A && B) || (!A && !B)\n\r");
    printf("0 | 0 | 1 | 1\n\r");
    printf("0 | 1 | 0 | 0\n\r");
    printf("1 | 0 | 0 | 0\n\r");
    printf("1 | 1 | 1 | 1\n\r");
    return 0;
}
```

Исходник приложу к домашнему заданию.

Задание 5.

The image displays two screenshots of a game interface titled "BooleanGame".

Top Screenshot:

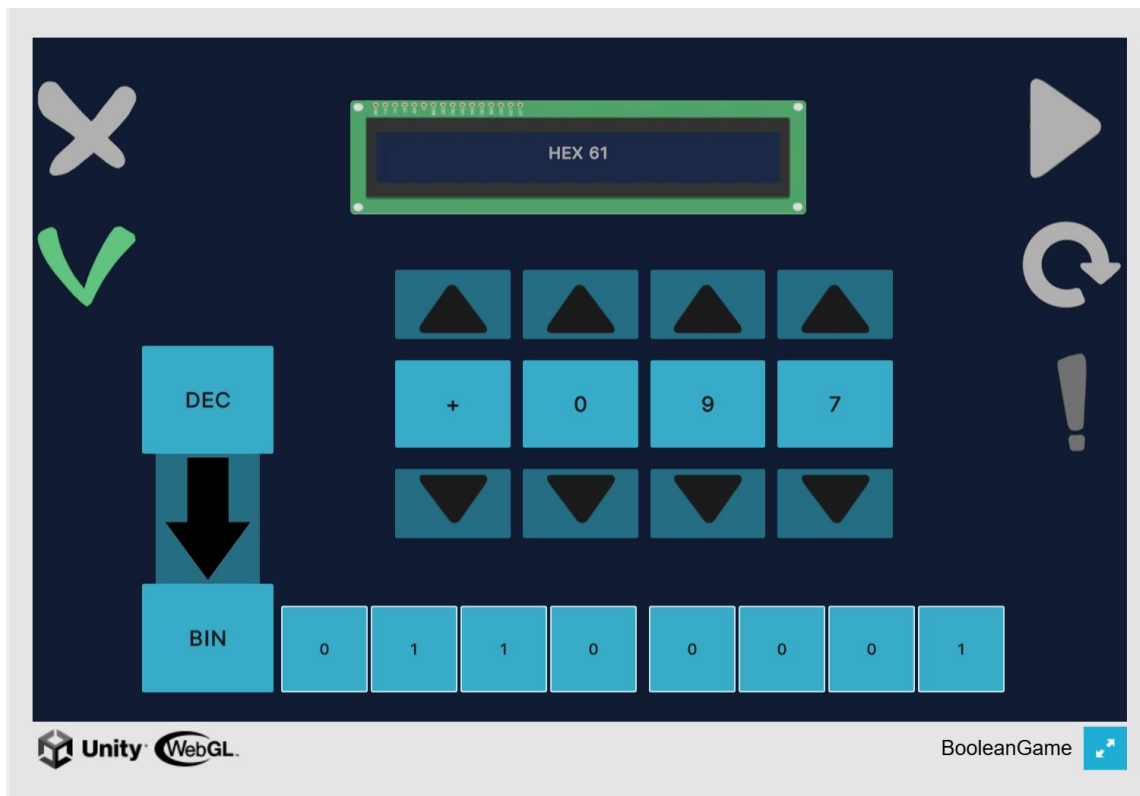
- Top center: A green-bordered display showing "DEC 50".
- Left side: A vertical stack with a "HEX" button, a large downward arrow, and a "BIN" button.
- Center: A 3x2 grid of buttons. The top row has two upward arrow buttons. The middle row has buttons with the numbers "3" and "2". The bottom row has two downward arrow buttons.
- Bottom center: A row of eight buttons containing the binary sequence: 0, 0, 1, 1, 0, 0, 1, 0.
- UI elements: A green checkmark on the left, a grey 'X' in the top left, and a play button, a refresh button, and an exclamation mark on the right.
- Footer: "Unity WebGL" logo on the left and "BooleanGame" with a share icon on the right.

Bottom Screenshot:

- Top center: A green-bordered display showing the equation $X = !A*B + A*!B + A*B$.
- Center: A 5x6 grid of buttons representing a truth table. The columns are labeled A, B, $!A*B$, $A*!B$, $A*B$, and X.
- Truth Table Data:

A	B	$!A*B$	$A*!B$	$A*B$	X
0	0	0	0	0	0
0	1	1	0	0	1
1	0	0	1	0	1
1	1	0	0	1	1

- UI elements: A green checkmark on the left, a grey 'X' in the top left, and a play button, a refresh button, and an exclamation mark on the right.
- Footer: "Unity WebGL" logo on the left and "BooleanGame" with a share icon on the right.



Задание 6. Упростить выражение $X = (B \rightarrow A) \cdot !(A + B) \cdot (A \rightarrow C)$

$$B \rightarrow A = !B + A;$$

$$A \rightarrow C = !A + C;$$

$$!(A + B) = !A * !B;$$

$$(!B + A) * (!A * !B) * (!A + C) = ((!B + A) * (!A * !B)) * (!A + C) ;$$

$$((!B + A) * (!A * !B)) = (!B * !A * !B + A * !A * !B) = !A * !B + 0 * !B = !A * !B;$$

$$(!A * !B) * (!A + C) = !A * !B * !A + !A * !B * C = (!A * !B) + (!A * !B) * C = !A * !B$$

Ответ: $!A * !B$